NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

B.Sc. – Zoology - Syllabus

(New Education Policy – 2020)

Offered by:

Department of Zoology

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

B.Sc. – Zoology – Syllabus

Year	Sem.	Course Code	Paper Title	Theory/ Practical	Hrs.	Credits
Certificate Course in Animal Diversity						
1	I	ZOOCC111T	Non-Chordata (Protozoa to Helminthes)	Theory	45	3
		ZOOCC112T	Non-Chordata (Annelida to Echinodermata)	Theory	45	3
	II	ZOOCC121T	Diversity and Biology of Chordata	Theory	45	3
		ZOOCC122P	Practical	Practical	45	3
Diploma in Applied Zoology and Animal Conservation						
2	III	ZOOCC231T	Environmental Biology and Wildlife	Theory	45	3
		ZOOCC232P	Practical	Practical	45	3
	IV	ZOOCC241T	Applied Zoology	Theory	45	3
		ZOOCC242T	Microbiology, Bioinformatics, Biostatistics, Biological Tools	Theory	45	3
Degree in Bachelor of Science						
3	V	ZOOCC351T	Animal Physiology and Biochemistry	Theory	60	4
		ZOOCC352T	Evolutionary Biology and Biosystematics	Theory	60	4
		ZOOCC353P	Practical	Practical	60	4
	VI	ZOOCC361T	Cytogenetics and Molecular Biology	Theory	60	4
		ZOOCC362T	Neurobiology, Physiological Constraints, Biotechnology, Toxicology	Theory	60	4
		ZOOCC363P	Practical	Practical	60	4

Maximum marks for theory and practical papers for each is 100 (70 external + 30 internal unit tests). Internal assessment includes presentation, projects, attendance, class performance and viva-voce.

(ZOO- Zoology, CC- Core Course, T- Theory, P- Practical)

B. Sc. in Zoology

Degree in Bachelor of Science

1. **APPLICABILITY**
   The NEP Regulations shall apply to the B.Sc. Zoology Programme from the session 2022-23.

2. **MINIMUM ELIGIBILITY FOR ADMISSION**
   Biology (Zoology and Botany) in Class XIIth.

Programme Objectives (POs):

Zoology as one of the subjects at undergraduate level, should be studied in an integrated and cross-disciplinary manner with a comprehensive understanding of all living systems and their relationship with the ecosystem. Within the broad-range skill sets related to the discipline, it is required to impart and assess the quality of critical thinking, analytical and scientific reasoning, and problem-solving capacity.

Programme Outcomes (POs):

- Develop deeper understanding of key concepts of Zoology at biochemical, molecular, cellular, physiological, histologicaland systematic level.
- Gain knowledge from classical descriptive to modern analytical disciplines of Zoology.
- Develop competency to make our students competent to excel in competitiveexaminations.
- Impart knowledge and skills through applied disciplines like Sericulture, Apiculture, Poultry and Aquaculture etc.
- Integrate and explore biological data. Use current laboratory setup, instrumentation, statistical and biological techniques in the collection, organization, analysis, interpretation and manipulating the data related to Zoology discipline and allied branches.
- Empower the students by enhancing their self-sustainability capabilities through a thorough understanding of skill-basedsubjects and techniques by learning.
- Develop social competence including listening, speaking, observational, effective interactive skills and presenting skills to meet global competencies.
- Understand the issues of environmental contexts andsustainable development.
- Aware students about ethical principles and commit toprofessional ethics and responsibilities.

B. Sc. I (Semesters I and II)

Programme Specific Outcomes (PSOs):

- Students will have a comprehensive knowledge of the Kingdom Animalia.
- Students will learn the distribution, diversity, classification, physiology, and form andfunction of each major animal lineage within Non-chordates and Chordates.
- Students will be able to apply fundamental principles of Zoology to make informed decisions on socio-scientific issues.
- Students will be able to apply for various positions in museums, wildlife/ biodiversity data collection, conservation programs, health care, and zoos etc. in both government and private labs/institutes including NGOs. The student will be offered 'CERTIFICATE COURSE IN ANIMAL DIVERSITY' after completing Ist year or two semesters.

B. Sc. II (Semesters III and IV)

Programme Specific Outcomes (PSOs):

- Students will be able to develop understanding of environmental conservation processes and its importance, pollution control and biodiversity and protection of endangeredspecies.
- The inclusion of environmental biology and wildlife will help students to understand the importance of the environment and how to conserve it.
- Students will understand the basic biology and life cycles of vectors, pests and parasites including epidemiology, diagnosis and treatment.
- At the end of the course the students will be able to comprehend the reason behind maintaining the equilibrium between flora and fauna on earth. They will be able to appreciate the environment and the interdependence between human, wildlife and nature for food production, maintaining clean air and water and sustaining biodiversity in a changing climate.
- Students will be able to analyze complex interactions among the various animals of different phyla, their distribution and their relationship with the environment.
- Students will gain knowledge of Agro based Small Scale Industries like sericulture, fish farming, apiculture, etc., which will help them in finding carrier opportunities.
- Students can get subsidy and loan from state government to start Poultry, Pisciculture and Apiculture, under various schemes run by state govt. and become "AATMNIRBHAR" and generate jobs for others.
- These Diploma courses will enable students to apply for various positions in museums, wildlife/biodiversity data collection, conservation programs, health care, and zoos etc. inboth government and private labs/institutes including NGOs as environmental consultants, manager, educator, outreach specialist, wildlife law enforcement officer, zoocurator, museum curator. Besides this, the students can also take up higher studies and research as their career. The student will be offered 'DIPLOMA IN APPLIED ZOOLOGY AND ANIMAL CONSERVATION' after completion of 2 years of the programme or 4 semesters.

B. Sc. III (Semesters V and VI)

Programme Specific Outcomes (PSOs):

- This programme aims to develop an understanding of structural, functional, biochemicaland behavioral aspects of life.
- The course in biosystematics is an integrative and unifying science and will help the students in studying the genotypic and phenotypic variation of species in relation to theenvironments in which they occur.
- This course will provide students with the basic knowledge of evolutionary biology, both presenting the general principles of the discipline and exploring in details theoreticalproblems and case studies.
- The students will understand the structure and function of the cell and the principles of genetics.
- The course will provide an insight into the life processes at the subcellular and molecular levels.
- This course will provide theoretical and applied knowledge on the effects of chemicalsubstances on human health.
- The principles of genetic engineering, gene cloning and related technologies will enablestudents to play an important role in applications of biotechnology in various fields.
- It will enhance basic knowledge about toxicants and daily stressors and the way to combat with them.
- After completion of 3 years of the programme or 6 semesters, the student will be offered the 'BACHELOR DEGREE IN ZOOLOGY'. This programme will enable students to go for higher studies like Masters and then pursue Ph.D. in Zoology and allied subjects.

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B.Sc. – Zoology (FIRST SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. I (Paper Code - ZOOCC111T)
NAME OF THE PAPER: Non-Chordata (Protozoa to Helminthes)

Objectives:
-To make the students aware of basic understanding of invertebrate animals.
-Facilitate the students to deep drive into studies of animal diversity.
-To develop basic understanding in students about evolutionary relationships among animal phyla.

Learning Outcomes:
At the completion, students will be able to:
-Identify and differentiate different lower invertebrate phyla and their associated animals.
-Explain basic biology and ecology of them.
-Understand economic importance of these lower invertebrates with reference to ecosystem and specially the man.

Unit I

Phylum Protozoa – characters, classification and types.
Paramecium – habit and habitat, structure, nutrition, digestion, excretion, respiration, osmoregulation, behavioral response, reproduction, binary fission, conjugation, autogamy, cytogamy, hemixis, endomixis.

Unit II

Phylum Porifera- characters, classification and types.
Sycon – Habit and habitat, external structure, canal system, body wall, nutrition, digestion, excretion, respiration, reproduction and development, types of water canal system.

Unit III

Phylum Coelenterata - characters, classification and types.
Obelia – Habit and habitat, morphology, polymorphism, development of medusa, physiology, reproduction, life cycle, alteration of generation and metagenesis.

Unit IV

Phylum Platyhelminthes – Characters, classification and types.
Fasciola –Habit and habitat, morphology, body wall, digestive system, excretory system, respiration, nervous system, reproduction, life history, types of larval stages, economic importance of *Fasciola*, parasitic adaptation of *Fasciola*.
Phylum Nemathelminthes – Characteristics, classification and types.
Ancylostoma – Habit and habitat, morphology, life cycle, disease caused, treatment and prevention.

Suggested Readings

1. Ruppert, EE, Fox R.S., Barnes R.D. (2004) *Invertebrate Zoology*, 7th Edition. Cengage Learning
2. Thomas Jeffrey Parker, William A. Haswell (2016). *Parker & Haswell's A Textbook of Zoology Volume 1*. WENTWORTH Press
3. Brusca (2016). *Invertebrates*. Sinauer
4. Pechenik Jan (2014). *Biology of the invertebrates*. McGraw Hill
5. Barnes R. S. K., Calow P. P., Olive P. J. W., Golding D. W., Spicer J. I. (2009). *The Invertebrates: A Synthesis*. Wiley Blackwell
6. Kotpal R.L. (2018) *Modern Text Book of Zoology: Invertebrates*. Rastogi Publications
7. Jordan E.L. and Verma P.S. : *Invertebrate Zoology*. S. Chand Publications.

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B.Sc. – Zoology (FIRST SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. II (Paper Code - ZOOCC112T)

NAME OF THE PAPER: Non-Chordata (Annelida to Echinodermata)

Objectives:

-To make the students understand evolutionary relationship among lower and higher groups of animals.
-To make the students learn about similarities and differences among different animal phyla.
-To develop in the students the understanding of structural and functional diversity.

Learning Outcome:

At the completion, the students will be able to:
-Correlate or differentiate various non-chordates on the basis of their anatomy and physiology.
-Know the hierarchy of development of animals from lower to higher level.
-Understand the life history and the developmental processes of animals.

Unit I

Phylum Annelida – Characters, classification and types.

Hirudinaria – Habit and Habitat, morphology, segmentation, receptors, sense organs, division of body, body apertures, body wall, locomotion, alimentary, digestive system, excretory system, respiratory system, nervous system, coelom and haemocoelomic system, reproduction, parasitic adaptations in *Hirudinaria*, economic importance of *Hirudinaria*.

Unit II

Phylum Arthropoda – Characters, classification and types.

Palaemon – Habit and habitat, morphology, exoskeleton, appendages, body wall, digestive system, excretory system, respiratory system, nervous system, sense organs or receptors, compound eyes, mosaic vision, apposition images and superposition images.

Unit III

Phylum Mollusca – Characters, classification and types.

Pila – Habit and Habitat, external features, body wall, mantle, digestive system, excretory system, respiratory system, mode of respiration, nervous system, blood vascular system, pearl formation in mollusca.

Unit IV

Phylum Echinodermata – Characters, classification and types.

Asterias - Habit and Habitat, external features, oral and aboral views, body wall, digestive system, excretory system, respiratory system, nervous system, water vascular system, larval stages in Echinodermata.

Suggested Readings

1. Ruppert, EE, Fox R.S., Barnes R.D. (2004) *Invertebrate Zoology*, 7th Edition. Cengage Learning
2. Thomas Jeffrey Parker, William A. Haswell (2016). *Parker & Haswell's A Textbook of Zoology Volume 1*. WENTWORTH Press
3. Brusca (2016). *Invertebrates*. Sinauer
4. Pechenik Jan (2014). *Biology of the invertebrates*. McGraw Hill
5. Barnes R. S. K., Calow P. P., Olive P. J. W., Golding D. W., Spicer J. I. (2009). *The Invertebrates: A Synthesis*. Wiley Blackwell
6. Kotpal R.L. (2018) *Modern Text Book of Zoology: Invertebrates*. Rastogi Publications
7. Jordan E.S. and Verma P.S. : *Invertebrate Zoology*. S. Chand Publications.

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B.Sc. – Zoology (SECOND SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. I (Paper Code - ZOOCC121T)
NAME OF THE PAPER: Diversity and Biology of Chordata

Objectives:

-To educate students about the comprehensive identification abilities of chordate diversity.

-To understand anatomical, physiological and behavioral differences among different vertebrate phyla.

Learning Outcome:

At the completion, students will be able to:

-Demonstrate comprehensive identification abilities of chordate diversity.

-Explain structural and functional diversity of chordates

-Explain evolutionary relationship amongst chordates

-Take up research in biological sciences.

Unit I

Hemichordata:

Balanoglossus – Systematic position, geographical distribution, external morphology, shape, size and colour, habit and habitat, division of body – proboscis, collar, trunk, body wall, coelom, endoskeleton, digestive system - (alimentary canal, food and feeding mechanism, digestion), respiratory system (branchial pharynx, branchial sac, mechanism of respiration), blood vascular system, excretory system, nervous system, reproductive system, affinities of *Balanoglossus*, Development – Fertilization, embryonic or prelarval development, larval development, Tornaria larva, metamorphosis.

Unit II

Urochordata:

Herdmania - Systematic position, geographical distribution, external morphology, shape, size and colour, habit and habitat, division of body – body proper, foot, test or tunic, spicules, mantle / body wall, internal organs / viscera, coelom and atrium, digestive system, alimentary canal, food and feeding mechanism, digestion, respiratory system, blood vascular system, course of blood circulation, excretory system, nervous system, neural complex, sense organs, reproductive system, affinities of Urochordata, Development of *Herdmania* – Fertilization, embryonic or prelarval development, larval development, retrogressive metamorphosis.

Unit III

Cephalochordata:

Amphioxus - Systematic position, geographical distribution, external morphology, shape, size and colour, habit and habitat, division of body, body apertures, body wall – skin, musculature, peritoneum, coelom, skeleton, atrium, digestive system (alimentary canal, food and feeding mechanism, digestive glands, physiology of digestion), respiratory system, blood vascular system, excretory system (Protonephridia, hatschek's nephridium, brown funnel, renal papillae), nervous system, sense organ / receptors, reproductive system.

Unit IV

Classification of different classes of vertebrates
Pisces, Amphibia, Reptilia, Aves, Mammalia up to order with characters and examples.

Suggested Reading

1. Young, J. Z. (2004). The Life of Vertebrates. III Edition. Oxford University Press.
2. Kenneth V. Kardong (2015). Vertebrates: Comparative Anatomy, Function, Evolution. McGraw Hill
3. Thomas Jeffrey Parker, William A. Haswell (2016) Parker & Haswell's A Textbook of Zoology Volume 2. WENTWORTH Press
4. Eroschenko, Victor P. (2008). diFiore's Atlas of Histology with Functional correlations. XII Edition. Lippincott W. & Wilkins

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B.Sc. – Zoology (SECOND SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. II (Paper Code - ZOOCC122P)
NAME OF THE PAPER: Practical

Objectives:

-This programme aims to introduce students to animal diversity of invertebrates. The students will be taught about invertebrates using observational strategies, museum specimens and field reports.

-Keeping in mind career perspectives in private and public labs, the lab courses have been substituted to ensure high quality training of students in laboratories.

Learning Outcome:

At the completion of the course students will learn:

-To use simple and compound microscopes.

-Different species of Invertebrates from Annelida followed by Echinodermata and more.

-The students will be having deep knowledge about their lifecycle, specific organs, genre, species etc.

1. Major dissection:
   i. Central Nervous System of *Palaemon*.
   ii. Central Nervous System of *Pila*.
   iii Nervous System of *Unio*
2. Minor dissection:
   i. Nerve ring of earthworm
3. Permanent stained preparation of:
   i. Gemmules of Sponges
   ii. Parapodium of *Neries*
   iii. Septal nephridia of Earthworm
4. Temporary / Glycerine preparation of:
   i. Spicules of sponges
   ii. Setae (in situ) of Earthworm
   iii. Statocyst of *Palaemon*.
   iv. Hastate plate of *Palaeomon*
   v. Radula of *Pila*
5. Spotting – Study of specimens and slides of Non-chordata (Protozoa, Porifera, Coelenterata, Platyhelminthes, Nemathelminthes, Annelida, Arthropoda, Mollusca, Echinodermata).
6. Project / Assignment
7. Class record
8. Viva-voce

Suggested Reading:

P. S. Verma: Practical Invertebrate Zoology
S. S. Lal : Practical Invertebrate Zoology

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B.Sc. – Zoology (THIRD SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. I (Paper Code - ZOOCC231T)

NAME OF THE PAPER: Environmental Biology and Wildlife

Objectives:

-Inclusion of ecology and environmental sciences will enrich students with our world which is crucial for human wellbeing and prosperity.

-To facilitate the understanding about environment, its components and importance.

-To make the students aware of global environmental issues, policies and practices.

-Learn, identify and differentiate between different categories of wild animals and their role to us.

Learning Outcome:

At the completion, students will:

-Know ecological basics and can contribute towards rich and healthy environment.

-Knowing the nature and natural resources, students will be able to practice the measures for their protection and conservation.

-Enable the student to understand, compare, think and evolve strategies for wildlife management

-Know the reasons behind vanishing wildlife and the way to overcome that.

Unit I

Ecology

Ecosystem – Concept, components, fundamental operations, energy, flow, food chain, food web, trophic levels, abiotic and biotic factors, ecological pyramids.

Ecological Niche – Introduction, examples of ecological niche, ecotypes and ecotones.

Ecological succession – Kinds of succession, patterns of succession – Xerosere, hydrosere, mesosere, theories of succession – Monoclimax theory, Polyclimax theory, Climax pattern theory.

Unit II

Concept, sources, types of air, water, soil, noise and radiation pollution. Control of environmental pollution. The Green house effect, Ozone depletion, Acid rain. Environmental awareness including water and resource conservation and sanitation.

Unit III

Wild life of India

Knowledge of endangered species (Reptiles, birds and mammals), causes of being endangered, economic importance. IUCN categories and Red Data Book.

Unit IV

Conservation of Wild Life

Important sanctuaries, National parks and Biosphere Reserves of India. Current status of different projects launched for the conservation of animal species, in-situ and ex-situ conservation of wild life.

Suggested Reading:

1. Odum E.P. (2005) *Fundamentals of Ecology*. Cengage Learning India Private Limited
2. Smith Thomas M., Smith Robert Leo (2014) *Elements of Ecology*. Pearson Education
3. Krebs, Charles J. 2009. *Ecology: the experimental analysis of distribution and abundance.* Pearson.
4. Sharma PD (2018). *Fundamentals of Ecology*. Rastogi Publications.
5. Sharma PD (2018). *Environmental Biology and Toxicology*. Rastogi Publications.
6. Pepper, I.L., Gerba, C.P. & Brsusseau, M.L. 2006. *Environmental & Pollution Science*. Elsevier Academic Press.
7. Gupta, K.R. 2006. *Environmental Legislation in India*. Atlantic Publishers and Distributors.

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B.Sc. – Zoology (THIRD SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. II (Paper Code - ZOOCC232P)
NAME OF THE PAPER: Practical

Objectives:

-To provide hands-on training to observe anatomical features of animals of economic importance.

-To provide knowledge of vertebrates through slides and museum specimens to enhance the student's understanding.

Learning Outcome:

At the completion of the course, students will be able to:

-Distinguish animals on the basis of their anatomical features.

-Identify the animals to their respective phyla on the basis of morphological features.

-Through histological and physiological tests, students can understand basic diagnostic tools used in forensic and diagnostic labs.

1. Major dissection:
   V^{th}, VIIth, IXth and X^{th} Cranial Nerves of *Scoliodon*
2. Minor dissection:
   i. Afferent branchial arteries of *Scoliodon*
   ii. Efferent branchial arteries of *Scoliodon*
   iii. Internal ear of *Scoliodon*
3. Permanent stained preparation of
   a) Placoid scales (Unstained)
   b) Ampullae of Lorenzini
4. Histology
   Stained preparation of striped / unstriped muscle fibres.
5. Comment upon physiological apparatus
   a) Kymograph
   b) Chromatography
6. Spotting – Study of speciemens and slides (Protochordata, Cyclostomata, Pisces, Amphibia, Reptilia, Aves, Mammalia), Models and Pictures of Endangered Animals.
   Osteology – Study of bones of Frog, Varanus, Fowl and Mammals
7. Project / Assignment
8. Class Record
9. Viva-voce

List of recommended books

P. S. Verma : Practical Vertebrate Zoology
S. S. Lal : Practical Vertebrate Zoology

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B.Sc. – Zoology (FOURTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. I (Paper Code - ZOOCC241T)
NAME OF THE PAPER: Applied Zoology

Objectives:

-To make students understand the economic loss caused by parasites, vectors and pests.
-To make students aware about different career prospective in possible associated and applicable fields.
-To make students understand the importance of wild life and its conservation.

Learning Outcome:

-After completion, students will be able to:
-Follow primary preventive and precautionary measures to obtain good health against vectors and parasites.
-Can manage pest population in agricultural and horticultural crops to increase crop productivity.
-Get employment in different applied sectors or can start their own business.
-Make significant contribution towards wildlife and its conservation.
-Enable students to take up research in Biological Science.

Unit I

Parasitology
Structure, life cycle, pathogenicity including diseases, causes, symptoms and control of the following parasites of domestic animals and humans :
Diphyllobothrium, Hymenolepis, Dracunculus, Wuchereria, Trypanosoma, Giardia, Paragonimus

Unit II

Vectors and pests
Vectors – Mosquito, housefly, bed bug, louse and their control.
Pest – types, characters, features, life cycle, nature of damage and control of termite, cockroach, cloth moth, wax moth, Gundhi bug, sugarcane leaf hopper.

Unit III

Apiculture, Sericulture, Lac culture

Unit IV

Aquaculture, Poultry, Vermiculture

Suggested Readings

1. Nigam H C (2014) Emerging Trends in Biology & Economic Zoology. Vishal PublishingCo.
2. Shukla GS & Upadhyay VB (2017) Economic Zoology Rastogi Publications
3. Srivastava KP and Dhaliwal GS. Textbook of Applied Entomology Volume 1 & 2. Kalyani Publishers.
4. Caughley, G., and Sinclair, A.R.E. (1994). Wildlife Ecology and Management. Blackwell Science.
5. Simpson: Principles of Animal Taxonomy (1962, Oxford).
6. Mayer & Ashlock: Principles of Systematic Zoology (2nd Edition, McGraw Hill).
7. Kapoor: Theory and Practicals of Animal Taxonomy (1988, Oxford & IBH).
8. Zar JH (2010) Biostatistical Analysis. 5th Edition. Pearson.

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B.Sc. – Zoology (FOURTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. II (Paper Code - ZOOCC242T)

NAME OF THE PAPER: Microbiology, Bioinformatics, Biostatistics, Biological Tools

Objectives:

-Make students aware of various industrial biological fields to which animals are associated.

-To know about basic scientific tools to manage the scientific data.

-Make students learn about working of biological tools essential in research.

Learning Outcome:

After completion, students will be able to:

-Distinguish among different useful and harmful microorganisms.

-Utilize the microorganisms against various diseases.

-Know about different tools, software and digital platforms for organizing and utilizing the biological data.

-Implement various analytical tools to evaluate and interpret the significance of the data.

-Choose their career in various industries accordingly.

Unit I

Microbiology – Distribution of microorganism in air, soil, water, milk, fruits and vegetables. Morphology, physiology and infection (outline) of bacterial and viral diseases, common antibiotic types, classification and mode of action.

Unit II

Bioinformatics:

Basic Scientific data resources (encyclopedia, biological abstracts, reviews, zoological records, current contents etc.). Storage, retrieval and citation of scientific data.

Unit III

Biostatistics

Sampling, measure of central tendency (Mean, median and mode) and dispersion (Variance, standard deviation and standard error), Test of significance of mean, the Chi-square test, correlation and regression.

Unit IV

Biological Tools

Principle and use of common physiological apparatus in electrophoresis, pH meter, Microscopy – Light, Scanning and transmission electron microscopes.

Suggested Readings

1. Microbiology: An Introduction: Tortora GJ, Funke BR, Case CL.
2. Prescott's Microbiology: Willey J, Sherwood L, Woolverton CJ.
3. Bioinformatics: Sequence And Genome Analysis by David W. Mount.
4. Introduction to Bioinformatics: Lesk AM.
5. Introduction to Biostatistics by Dr. Pranab Kr. Banarjee
6. Biostatistics: Triola MM and Triola MF.
7. Bioinstrumentation by L. Veerakumari
8. Bioinstrumentation: Webster JG.

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B.Sc. – Zoology (FIFTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. I (Paper Code - ZOOCC351T)
NAME OF THE PAPER: Animal Physiology and Biochemistry

Objectives:

-To make the student aware of functioning of the body and its various organs.
-To develop in the student an understanding of the various homeostatic systems of the body.
-To develop in the student an understanding of regulation of function in the body
-To know the basic need of nutrients for growth and development of the body.

Learning Outcome:

After completion, students will be able to:
-Understand the mechanism underlying maintenance of homeostasis of the body.
-Have enhanced knowledge about human physiological systems.
-Understand how these separate systems interact to yield integrated physiological responses.
-Understand structure and function of biomolecules.
-Explain Enzyme catalysis and kinetics

Unit I

Digestion and absorption – Definition, types of digestion – intracellular digestion, extracellular digestion; Structure of alimentary canal and associated digestive glands (liver, pancreas, gall bladder); Digestion in mouth, digestion in stomach, digestion in intestine, defaecation; Absorption of carbohydrates, proteins, fats.

Respiration – Introduction, external respiration, breathing, process of gaseous exchange in the air and blood during respiration, condition of gases in the blood, respiratory exchange in tissues, transportation of oxygen and carbon dioxide, carbon dioxide dissociation curve, tissue respiration, regulation of respiration, chloride shift / hamburger phenomenon.

Unit II

Circulatory system – Introduction, structure of mammalian heart. Blood vascular system, arterial and venous system, heartbeat, functioning of heart, cardiac cycle, regulation of heart beat, heart sounds – lupp, dupp, pulse, blood pressure.

Excretion – Introduction, structure of kidney, functions of kidney – Urine formation, glomerular filtration, selective reabsorption, tubular secretion, counter current mechanism, regulation of kidney function, control of kidney function, urea formation – ornithine cycle.

Unit III

Endocrine glands – Introduction, nature of endocrine glands, pituitary gland, thyroid gland, parathyroid gland, adrenal gland, thymus gland, pineal body.

Unit IV

Biochemistry – General Chemistry and classification for carbohydrates, lipids and proteins.

Enzymes – Introduction, exoenzyme, endoenzyme, classification, nomenclature, chemical nature of enzyme, mechanism of enzyme action, activation of enzyme, isoenzyme, coenzyme, holoenzyme, apoenzyme, cofactor, factor influencing enzyme action. Enzyme inhibitors- Reversible and irreversible, competitive and noncompetitive.

Suggested Reading

1. Guyton, A.C. & Hall, J.E. (2006). Textbook of Medical Physiology. XI Edition. Hercourt Asia PTE Ltd. /W.B. Saunders Company.
2. Tortora, G.J. & Grabowski, S. (2006). Principles of Anatomy & Physiology. XI Edition John Wiley & sons
3. Christopher D. Moyes, Patricia M. Schulte 2016 Principles of Animal Physiology. 3rd Edition, Pearson Education,
4. Ganong: Review of Medical Physiology, Lang Medical Publ.
5. Boyer: Concepts in Biochemistry (3rd ed. 2006, Brooks/Cole)

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B.Sc. – Zoology (FIFTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. II (Paper Code - ZOOCC352T)
NAME OF THE PAPER: Evolutionary Biology and Biosystematics

Objectives:

-To make students know about the origin of life on earth and its possible extraterrestrial existence.

-To enhance the knowledge about various sources of variations and their role in evolution.

-Understanding of animal taxonomy and systematic and their application

-Enhance the knowledge about molecular basis of animal taxonomy.

Learning Outcome

At completion, students will be able to:

-Understand the concept, process and patterns of evolution.

-Acquire knowledge and reasoning skills useful to interpret biological phenomena in the light ofevolution.

- Understand about how to position any animal in a classification system.

-Conceptualize different theories of biosystematics to implement accordingly in the required field.

Unit I

Evolutionary Biology: Historical background, evolution, old and new theories of origin of life; Biochemical origin of life, extra-terrestrial life.

Unit II

Evidences of Organic Evolution

Evidences from classification, evidences from comparative anatomy, evidences from embryology, evidences from vestigial organs, evidences from physiology and Biochemistry, evidences from enzymes and hormones, evidences from cytology, evidences from connecting links, evidences from zoogeography, evidences from genetics.

Theories of Evolution

Lamarkism, Darwinism and Modern synthetic theory.

Unit III

Introduction to taxonomy and biosystematics:

Definition and basic concepts of taxonomy and systematics, Types and operation of taxonomy,

Importance of taxonomy and biosystematics, International Code of Zoological Nomenclature (ICZN), Aims and tasks of a taxonomist, Animal diversity: Alpha, Beta and Gamma

Unit IV

Biological Classification:

Kinds of Zoological classification, Taxonomic (Linnean) hierarchy, Concept of species, Mechanism of speciation, Theories of biological classification, Kinds and components of classification.

Suggested Reading

1. Futuyma, Douglas J. and Kirkpatrick Mark. Evolution (4th Edition) Sinauer
2. Veer Bala Rastogi (2017) Organic Evolution. Med Tech
3. Hall B.K. and Hallgrimsson B. (2008). *Strickberger's Evolution*. IV Edition. Jones and Bartlett Publishers Inc.
4. Huxley Julian. Evolution: The Modern Synthesis. Harper and Brothers.
5. Kapoor V.C. 1991.Theory and Practice of Animal Taxonomy. Oxford and IBH Publishing

NATIONAL P.G. COLLEGE, LUCKNOW

(An Autonomous College of University of Lucknow)

B.Sc. – Zoology (FIFTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. III (Paper Code - ZOOCC353P)
NAME OF THE PAPER: Practical

Objectives:

-Provide students hands-on training on the techniques like microscopy, centrifugation, chromatography, DNA isolation and various other biochemical techniques.

-To make students aware of the animals of major economic importance.

Learning Outcome

-Students can learn about the structure and function of DNA.

-They can know how to estimate blood group.

-This will help students in getting basic working of pathological labs and also help in getting employment within them to contribute in health care system.

1. Major Dissection
   Central nervous System of Cockroach
2. Minor Dissection
   a) Salivary Gland complex of Cockroach
   b) Mouth Parts of Cockroach
3. Immunology
   A. Detection of blood group.
4. Physiology
   a) Preparation of Haemin crystals.
   b) Preparation of vertebrate blood film stained with Leishman's stain.
   c) Detection of sugar / albumin / ketone bodies in an artificial sample of urine.
5. Permanent preparation
   1. Cimex
   2. Pediculus
   3. Haematopinus
   4. Scales of Fish
6. Temporary Preparation
   1. Mouth parts of mosquito
   2. Mouth parts of housefly
   3. Redia Larva
   4. Cercaria Larva
7. Comment upon physiological apparatus:
   1. Haemocytometer
   2. Haemoglobinometer
   3. pH meter
8. Spotting-
   Study of speciemens and slides of Protozoan parasites, Helminths, Arthropods, nematodes and edible fishes. Prepared slides of Histology and Embryology.
9. Assignment / Project
10. Class Record
11. Viva-voce

List of recommended books

P. S. Verma : Practical Vertebrate Zoology
S. S. Lal : Practical Vertebrate Zoology

NATIONAL P.G. COLLEGE, LUCKNOW

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B.Sc. – Zoology (SIXTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. I (Paper Code - ZOOCC361T)
NAME OF THE PAPER: Cytogenetics and Molecular Biology

Objectives:
-To understand the structure and functions of cell and its organelles.
-To understand the structure and arrangement of genes and their interaction.
-To understand the structure and types of DNA and its organization.
-Enhance the knowledge about the life processes at cellular and molecular levels.

Learning Outcome:
At completion, students will be able to:
-Understand the structure and function of the cell organelles and process of cell division.
-Understand the process through which information is inherited from parent to progeny.
-Develop an understanding of concepts, mechanisms and evolutionary significance and relevance of molecular biology in the current scenario.

Unit I

Structure, types and function of cell and cell organelles (Plasma membrane, Mitochondria, Nucleus, Endoplasmic reticulum, Golgi Complex, ribosome and lysosomes)

Unit II

Chromosomes – Types of chromosomes, structure of chromosomes, chemical composition of chromosomes, functions of chromosomes, special type of chromosomes – Lampbrush Chromosomes, Polytene chromosomes.
Watson and Crick model of DNA, differences between DNA and RNA.

Unit III

Mendel's law of inheritance – Monohybrid cross, Dihybrid cross, Test Cross, Back cross, Incomplete dominance.
Linkage – Introduction, coupling and Repulsion Hypothesis, examples of linkage, types of linkage, arrangement of linked genes, Chromosome theory of linkage, significance of Linkage.
Crossing Over – Introduction, examples of crossing over, mechanism of crossing over, kinds of crossing over, percentage of crossing over, factors affecting crossing over, utility of crossing over.
Sex Determination – genetically controlled sex determination, environmental role in sex determination, hormonal role in sex determination, chromosomal theory of sex determination, XX-Xy type / lygaeus type, XX-XO type / Protenor type. Quantitative theory of sex determination, genic balance theory, cytoplasmic sex determination, barr body / Lyons hypothesis.
Hereditary diseases of men

Unit IV

Nucleic Acids: structure, replication, central dogma, genetic code
Protein synthesis (Transcription, Translation)
RNA processing
Gene regulation

Suggested Reading

1. Karp, G. (2010). *Cell and Molecular Biology: Concepts and Experiments*. VI Edition. John Wiley and Sons. Inc.
2. De Robertis, E.D.P. and De Robertis, E.M.F. (2006). *Cell and Molecular Biology*. VIII Edition. Lippincott Williams and Wilkins, Philadelphia.
3. Cooper, G.M. and Hausman, R.E. (2009). *The Cell: A Molecular Approach*. V Edition. ASM Press and Sunderland, Washington, D.C.; Sinauer Associates, MA.
4. Gardner, E.J., Simmons, M.J., Snustad, D.P. (2008). *Principles of Genetics*. VIII Edition. Wiley India.

NATIONAL P.G. COLLEGE, LUCKNOW

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B.Sc. – Zoology (SIXTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. II (Paper Code - ZOOCC362T)
NAME OF THE PAPER: Neurobiology, Physiological Constraints, Biotechnology, Toxicology

Objectives:
-To enhance the knowledge about neuronal conduction.
-To get aware of psychophysiological changes in body during unfavorable conditions.
-To develop an understanding about genetic engineering and its importance.
-To make students understand about toxicants and their effects.

Learning Outcome:
On completion, students will be able to:
-Understand basic mechanism of nerve impulse conduction.
-Cope up with stressful events and to overcome from it.
-Implement the significance of Biotechnology in applicable fields.
-Understand the mechanism, effects and solutions of toxicants.

Unit I
Neurobiology
Neurons and glia and their types, synapse, membrane and action potentials, synaptic potential, synaptic integration, propagation and conduction of nerve impulse, synaptic transmission, electrical and chemical transmission.

Unit II
Physiological constraints:
Kinds of stress and their influence, physiological mechanism of stress. Neuroendocrine regulation of stress.
Effect of weightlessness (In space environment) on neurosensory, cardiovascular and respiratory systems.

Unit III
Biotechnology – Definition, Genetic engineering, concept and recombinant DNA technology and its application in agriculture, medical areas and energy production. Biotechnology of food processing, Pharmaceuticals, uses of microbes in insulin production and fermentation.

Unit IV
Toxicology
Exposure of toxicants, routes of exposure, duration and frequency of exposure, dose response relationship, categories of toxic effects, local and systematic effects, immediate and delayed effects, reversible and irreversible effects.

Suggested Reading

1. Principles of Neural Science: Kandel ER, Schwartz J, Jessell T.
2. Fundamental Neuroscience: Squire LR, Bloom FE, Splitzer NC
3. Biotechnology: RC Dubey. S. Chand Publications
4. Genetic Engineering: TA Brown.
5. Stress Physiology in Animals: Balm PHM. Sheffield Academic.
6. Handbook of Toxicology, M.J. Derelanko & C.S. Auletta, 3rd Ed. CRC Press
7. Concepts of Toxicology. Omkar

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B.Sc. – Zoology (SIXTH SEMESTER)– Syllabus (Maximum Marks=70)

PAPER NO. III (Paper Code - ZOOCC363P)
NAME OF THE PAPER: Practical

Objectives:
Provide students the basic understanding of anatomy of vertebrate (fish) and certain hands on exposure to different techniques associated with genetic material.

Learning Outcome:
At the completion, students will be able to know physiology of nervous system. They will develop keen understanding about different pest, vectors and parasites. They will know the functionality of biological tools and techniques.

1. Dissection –
   a. Take out Afferent branchial arteries, efferent branchial arteries, cranial nerves (V^{th}, VIIth, IXth and X^{th} and their branches) of *Wallago*.
   b. Central nervous system of *Periplaneta americana* (Cockroach).
2. Minor dissection –
   a. Salivary gland complex of Cockroach
   b. Webberian ossicle of *Wallago*.
3. Culture and gram staining of Bacteria.
4. Preparation of:
   a. Slide of nerve cells from given samples.
   b. Cheek epithelial cells with Methylene Blue.
5. Cytological preparation- Make a slide of salivary gland for giant chromosomes of *Chironomous / Drosophila* larva.
6. Molecular Biology –
   A. Isolation of genomic DNA from Animal cell.
   B. Electrophoresis of DNA samples.
7. Spotting – Animal's Adaptations, hazardous effects of pollution / Toxicology, Behavioural models, Pest/Vectors/ Parasites. Life cycles.
8. Comment upon the biological tools
   a. Autoclave
   b. Centrifuge
   c. Colorimeter
9. Assignment / project
10. Class Record
11. Viva-voce

List of recommended books

P. S. Verma : Practical Vertebrate Zoology
S. S. Lal : Practical Vertebrate Zoology